Problem Set 22

Volumes III: Known Cross-Sections

October 29, 2026

Problem 3

$$\begin{aligned} V &= 4\int_{-1}^{1} \frac{\sqrt{3}}{4}(1-x^2)\,dx \\ &= \sqrt{3}(x - \frac{x^3}{3})\Big|_{-1}^{1} \\ &= \boxed{\frac{4\sqrt{3}}{3}} \end{aligned}$$

Problem 4

(b)

$$\begin{aligned} V &= \int_{0}^{1} x^2\,dx \\ &= \frac{x^3}{3}\Big|_0^1 = \boxed{\frac{1}{3}} \end{aligned}$$

(c)

$$\begin{aligned} V &= \int_{-1}^{1} \frac{1}{2}(1-x^2)^2\,dx \\ &= \frac{1}{2}\int_{-1}^{1} x^4 - 2x^2 + 1\,dx \\ &= \frac{1}{2}(\frac{x^5}{5} - \frac{2x^3}{3} + x)\Big|_{-1}^{1} \\ &= \boxed{\frac{8}{15}} \end{aligned}$$

(e)

$$\begin{aligned} V &= \int_{0}^{3} \frac{1}{2}\cdot\frac{5}{3}x\cdot\frac{4}{3}x\,dx \\ &= \frac{10}{9}\cdot\frac{x^3}{3}\Big|_0^3 \\ &= \boxed{10} \end{aligned}$$

Problem 5

(a)

$$\begin{aligned} V &= \int_0^h \frac{h^2}{L^2} \cdot y^2 \, dy \\ &= \frac{L^2}{h^2}(\frac{y^3}{3})\Big|_0^h \\ &= \boxed{\frac{1}{3}L^2h} \end{aligned}$$

(b)

$$\begin{aligned} V &= \int_0^h \frac{r^2}{h^2} \cdot \pi y^2 \, dy \\ &= \frac{\pi r^2}{h^2}(\frac{y^3}{3})\Big|_0^h \\ &= \boxed{\frac{1}{3}\pi r^2h} \end{aligned}$$

(c)

$$V = \int A(y) \, dy$$

The precise location of every single layer may shift, as long as the area of each piece remains the same.

(d)

$$\begin{aligned} A_1 &= \pi(r^2 - y^2) \\ A_2 &= \pi r^2 - \pi y^2 \\ A_1 &= A_2 \\ V_{sphere} &= \frac{1}{3}\pi r^2h + \pi r^2h = \boxed{\frac{4}{3}\pi r^2h} \end{aligned}$$

Problem 6

(b)

$$\begin{aligned} r_1 = 1.5^2 + 2.2^2 + 3.7^2 + 3.2^2), r_2 = 2.2, r_3 = 3.7, r_4 = 3.2 \\ V = 2\pi(1.5^2 + 2.2^2 + 3.7^2 + 3.2^2) = \boxed{62.04\pi} \end{aligned}$$

Problem 7

(a)

$$V = \frac{\pi}{8}\int_0^\pi \sin^2 x \, dx$$

$$= \frac{\pi}{8} \int_0^{\pi} \frac{1 - \cos 2x}{2} \, dx$$
$$= \frac{\pi}{8} (\frac{x}{2} - \frac{\sin 2x}{4}) \Big|_0^{\pi}$$
$$= \boxed{\frac{\pi^2}{16}}$$

(b)

$$V = \int_0^1 \frac{1}{1 + x^2} \, dx$$
$$= \arctan x \Big|_0^1$$
$$= \boxed{\frac{\pi}{4}}$$

(c)

$$V = \frac{\sqrt{3}}{4} \int_1^e \ln^2 x \, dx$$
$$= \begin{bmatrix} u = \ln^2 x & dv = dx \\ du = \frac{2 \ln x}{x} \, dx & v = x \end{bmatrix} \frac{\sqrt{3}}{4} (x \ln^2 x \Big|_1^e - \int_1^e 2 \ln x \, dx)$$
$$= \frac{\sqrt{3}}{4} e + \frac{\sqrt{3}}{2} (x \ln x - x) \Big|_e^1$$
$$= \boxed{\frac{\sqrt{3}}{4} (e - 2)}$$

Problem 8

(a)

$$R = \int_0^4 \sqrt{x} - \frac{x}{2} \, dx$$
$$= (\frac{2}{3} x^{\frac{3}{2}} - \frac{x^2}{4}) \Big|_0^4$$
$$= \boxed{\frac{4}{3}}$$

(b)

$$V = \int_0^4 (\sqrt{x} - \frac{x}{2})^2 \, dx$$
$$= \int_0^4 \frac{x^2}{4} - x^{\frac{3}{2}} + x \, dx$$

$$= (\frac{x^3}{12} - \frac{2}{5}x^{\frac{5}{2}} + \frac{x^2}{2})\Big|_0^4$$
$$= \boxed{\frac{8}{15}}$$

(c)

$$V = \frac{\pi}{8} \cdot \frac{8}{15} = \boxed{\frac{\pi}{15}}$$

(d)

$$V = \int_0^2 (2y - y^2)^2 \, dy$$
$$= \int_0^2 y^4 - 4y^3 + 4y^2 \, dy$$
$$= (\frac{y^5}{5} - y^4 + \frac{4y^3}{3})\Big|_0^2$$
$$= \boxed{\frac{16}{15}}$$

(e)

$$V = \int_0^4 \frac{x^2}{4} - x^{\frac{3}{2}} + x \, dx$$
$$\int_0^c \frac{x^2}{4} - x^{\frac{3}{2}} + x \, dx = \frac{4}{15}$$

The volume is strictly increasing on the interval $(0, 4)$.

Problem 9

(a)

$$2\cos x - \sec x$$
$$= 2\cos x - \frac{1}{\cos x}$$
$$= \frac{2\cos^2 x - 1}{\cos x}$$
$$= \frac{2\cos x}{\cos x} \geq 0$$
$$s(x) = 2\cos x - \sec x$$
$$s(x)^2 = 4\cos^2 x - 4 + \sec^2 x$$

(b)

$$\begin{aligned} V &= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} 4\cos^2 x - 4 + \sec^2 x \, dx \\ &= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} 2\cos 2x - 2 + \sec^2 x \, dx \\ &= (\sin 2x - 2x + \tan x)\Big|_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \\ &= 2(1 - \frac{\pi}{2} + 1) \\ &= \boxed{4 - \pi} \end{aligned}$$

(c)

$$V = \frac{\sqrt{3}}{4}(4 - \pi) = \boxed{\sqrt{3} - \frac{\sqrt{3}}{4}\pi}$$

(d)

$$\begin{aligned} s^2(\frac{\pi}{8}) &= 2 + \sqrt{2} - 4 + 4 - 2\sqrt{2} = 2 - \sqrt{2} \\ V &= 2 \cdot \frac{\pi}{4} \cdot (2 - \sqrt{2}) = \boxed{\frac{\pi}{2}(2 - \sqrt{2})} \end{aligned}$$

The Midpoint Rule overestimates the real value.

Problem 10

(a)

$$\begin{aligned} V &= \int_{-r}^{r} \frac{1}{2}(r^2 - x^2) \, dx \\ &= \frac{1}{2}(r^2 x - \frac{x^3}{3})\Big|_{-r}^{r} \\ &= \boxed{\frac{2}{3}r^3} \end{aligned}$$

(b)

$$\begin{aligned} V &= 4\int_{-r}^{r} r^2 - z^2 \, dz \\ &= 4(r^2 z - \frac{z^3}{3})\Big|_{-r}^{r} \\ &= \boxed{\frac{16}{3}r^3} \end{aligned}$$

(c)

$$\begin{aligned} A_1 &= 4r^2 - 4z^2 \\ A_2 &= \pi r^2 - \pi z^2 \\ V &= \int_{-r}^{r} \pi(r^2 - z^2)\, dz \\ &= \boxed{\frac{4}{3}\pi r^3} \end{aligned}$$

(d) The π disappears as a result of the area; all the radicals disappear, leaving a simple polynomial behind.